

Midas_V2 – v0.3.12 Documentation

Filters Roadmap (China Exclusion, Catalyst Requirement, Market Context Logging)

Version Context

- **Previous tag:** v0.3.11 (pre-flight project guard + docs).
- **This tag:** v0.3.12-filters-roadmap.
- **Purpose:** Add optional filters (China exclusion, catalyst requirement) and logging (market context) without breaking existing scenario logic.

A. Principles & Scope

- **Keep Scenario logic intact** (A–E). Filters live *before* strategy triggers.
- **No breaking changes**: default behavior remains identical with flags **off**.
- **Minimal new files**; prefer editing existing scripts.
- **One-liners only** for run commands.

B. Config Flags (add to `config/settings.json` or `scenarios.json`)

Add a new, global `filters` block and per-scenario overrides optional.

```
```json
{
 "filters": {
 "exclude_china": true,
 "require_catalyst": true,
 "catalyst_window_minutes": 240,
 "market_ctx": {
 "enable": true,
 "symbol": "SPY",
 "gate": false,
 "gate_min_change_pct": 0.60,
 "gate_window_minutes": 30
 }
 }
}
```

```
}
}
...
```

#### **\*\*Notes\*\***

- ``exclude_china=true`` drops CN issuers from the `*universe*` before backtesting.
- ``require_catalyst=true`` keeps only symbols with a recognized news catalyst within the window prior to first trade gate (e.g., 4h pre-`open` → gate time).
- ``market_ctx.enable=true`` merely logs the context; ``gate=false`` means “do not filter yet.” Turn on later if experiments show benefit.

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## **C. Data Plumbing (minimal, additive)**

### 1. **\*\*Universe with metadata\*\***

- Extend ``scripts/topgappers.py`` to output ``data/universe/universe_YYYY-MM-DD.csv`` with columns:  
``date,symbol,price,gap_pct``
- **\*\*new:\*\*** ``country, is_china, has_catalyst, catalyst_source, catalyst_type, catalyst_time``
- Keep the plain ``universe_sample.txt`` for backward compatibility.

### 2. **\*\*Backtester intake\*\***

- In ``midas_v2/loader/universe.py`` (or equivalent), if the CSV exists, join metadata into the per-symbol run context; else fall back to TXT.

### 3. **\*\*Run-day script\*\*** (``scripts/run_day_simple.py``)

- Add ``--use-universe-csv`` (default **\*\*on\*\*** if file exists).
- Respect ``filters.exclude_china`` and ``filters.require_catalyst`` when constructing the working symbol list.

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## **D. China Filter (deterministic, simple)**

#### **\*\*Strategy\*\*** (fastest path):

1. Use issuer country / country of incorporation from your **\*\*reference endpoint\*\*** (Polygon ``v3/reference/tickers``, cached locally).
2. Mark ``is_china = (country in {CN}) OR (exchange in {HKEX} ADR→CN)``.
3. Maintain a small **\*\*denylist\*\*** for edge cases (e.g., shell ADRs): ``config/denylist_china.txt`` (optional).

#### **\*\*Implementation\*\***

- Edit `scripts/topgappers.py`:
- When building the daily universe, query/cached read reference metadata → set `country` and `is\_china`.
- If `exclude\_china`, drop those rows before writing `universe\_YYYY-MM-DD.csv`.
- Caching: write `data/ref/ticker\_meta.parquet` keyed by `symbol` with `country`, exchange, last\_updated`.

#### **\*\*Acceptance\*\***

- For a known CN small cap, confirm it is excluded when the flag is on and included when off.

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## **E. Catalyst Filter (modular, keyword-based first; API later)**

#### **\*\*Two-phase plan\*\***

##### **1. \*\*Phase 1 – Heuristic/Keyword (no new subscription requirements)\*\***

- Parse pre-market PR/news text from a lightweight source you already have or from locally stored notes.
- Accept keywords: `[earnings, 'FDA', 'phase', 'trial', 'approval', 'agreement', 'contract', 'guidance', 'merger', 'acquisition', 'partnership', 'licensing', 'ticker mentions + %']`.
- Time gate: within `catalyst\_window\_minutes` before the strategy gate.
- Store `has\_catalyst (bool)`, `catalyst\_type`, `catalyst\_time`, `catalyst\_source` in universe CSV.

##### **2. \*\*Phase 2 – API-based (Polygon News, Finnhub, Benzinga, etc.)\*\***

- Add `scripts/fetch\_news.py --date YYYY-MM-DD --symbols` that populates `data/news/YYYYMMDD.json`.
- Simple scorer selects the top most relevant headline and sets `has\_catalyst` + fields above.

#### **\*\*Implementation (Phase 1, minimal changes)\*\***

- Extend `scripts/topgappers.py` to call a small helper `src/midas\_v2/services/catalyst.py`:
- `infer\_catalyst(symbol, date) -> (has\_catalyst, type, ts, source)`
- Start with a stub that reads from `data/manual\_catalysts/YYYYMMDD.csv` (manual tagging allowed). This lets you backfill past days rapidly.
- Later swap stub with API calls without changing the call site.

#### **\*\*Backtest filter\*\***

- If `require\_catalyst=true`, keep only rows where `has\_catalyst==True`.

#### **\*\*Acceptance\*\***

- Manually tag 2–3 symbols for 2025-08-05; confirm only tagged ones are traded when `require\_catalyst=true`.

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## F. Market Context (log now, experiment later)

**\*\*What to log\*\***

- `SPY` (or `DIA`) % change from prior close to: (a) open, (b) gate time, (c) 30min window high/low.
- Simple trend label at gate: `UP` if price > VWAP & 9EMA on 1min; else `DOWN`.

**\*\*Implementation\*\***

- In `scripts/run\_day\_simple.py` before processing symbols:
- Load SPY minute bars for the date (you already fetch minutes).
- Compute `pct\_from\_prev\_close`, `intraday\_change\_0\_30m`, `vwap\_relation`.
- Persist to `out//market\_ctx.json` and attach to run summary.

**\*\*Optional gate (off by default)\*\***

- If `filters.market\_ctx.gate=true`, then proceed only if `intraday\_change\_0\_30m >= gate\_min\_change\_pct`.

**\*\*Acceptance\*\***

- Verify JSON appears and contains fields; when `gate=true`, confirm symbols are skipped if the condition fails.

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## G. CLI Additions (one liners)

1. **\*\*Single day (auto universe, filters applied)\*\***

...

```
python scripts/run_day_simple.py --date 2025-08-05 --scenario D
```

...

2. **\*\*Range run with summaries (filters respected)\*\***

...

```
python scripts/run_range_and_summarize.py --start 2025-08-05 --end 2025-08-31 --scenario D
```

...

3. **\*\*Rebuild summaries after toggling flags\*\***

...

```
python scripts/rebuild_range_summary.py --start 2025-08-05 --end 2025-08-31 --scenario D
```

...

\*No extra flags needed; the scripts read `filters` from config and the universe CSV if present.\*

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## H. Metrics & Reporting

- Add columns to daily results summary:
- ``n_universe_pre`, `n_after_china_filter`, `n_after_catalyst_filter``.
- Market context snapshot: ``spy_gate_pct`, `spy_trend_gate``.
- Compare WR/PnL with a 2×2 matrix over a month:
- Baseline vs ``exclude_china``
- Baseline vs ``require_catalyst``
- Target: **\*\*WR +3–8 pts\*\*** from China exclusion; **\*\*WR +5–12 pts\*\*** from Catalyst (historically typical in similar projects).

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## I. Minimal Code Changes (file■by■file)

- ``scripts/topgappers.py``
- + reference lookup (country/exchange) with small local cache
- + write ``universe_YYYY-MM-DD.csv`` with metadata columns
- + stubbed ``infer_catalyst()`` call (manual file first)
- ``scripts/run_day_simple.py``
- + detect CSV universe and read metadata
- + apply filters based on config
- + compute market context JSON (always log)
- ``midas_v2/loader/universe.py`` (or similar intake)
- + merge metadata into symbol context; expose ``ctx.is_china`, `ctx.has_catalyst``
- ``config/settings.json`` (or ``scenarios.json``)
- + ``filters`` block shown above
- ``tools/`` (optional)
- ``services/catalyst.py`` (stub now → API later)

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## J. Backfill & Manual Tagging Workflow (fast start)

1. Create ``data/manual_catalysts/20250805.csv`` with headers:
  - ``symbol,catalyst_type,catalyst_time,catalyst_source``

2. Rerun ``run_day_simple.py``; the stub marks ``has_catalyst`` from this file.
3. Iterate a few days to validate lift before wiring an external News API.

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## K. Versioning & Guardrails

1. Commit and tag this change set:

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```
git add -A && git commit -m "v0.3.12: China exclusion + catalyst filter (stub) + market context logging"
&& git tag -a v0.3.12 -m "v0.3.12: Filters roadmap implemented" && git push && git push --tags
```

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2. If anything regresses:

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```
git reset --hard v0.3.11 && git push --force-with-lease
```

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## L. Validation Plan (7day pilot)

1. Run `**D_strict**` for 2025-08-05 → 2025-08-31 with ``exclude_china=true``, ``require_catalyst=false``. Capture lift.
2. Repeat with ``require_catalyst=true`` using manual tags for top movers only.
3. Compare WR/PnL; if positive, keep flags on and proceed to API integration for catalysts.

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## M. Future (API Catalyst, Better Features)

- Plug Polygon/Benzinga/Finnhub news → replace manual CSV.
- Add `**"quality score"**` for catalysts (e.g., FDA > earnings > PR fluff) to scale risk sizing later.
- Consider `**sector heat**` and `**opening RVOL gate**` once filters stabilize.

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## N. Quick ToDo Checklist

- [ ] Add ``filters`` block to config
- [ ] Patch ``topgappers.py`` to emit universe CSV with metadata

- [ ] Implement stubbed `infer\_catalyst()` backed by `data/manual\_catalysts/`
- [ ] Modify `run\_day\_simple.py` to read CSV + apply filters
- [ ] Log `market\_ctx.json` per day
- [ ] Run 7-day pilot; compare WR/PnL
- [ ] Tag results and decide on API integration